

Numerical Integration

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Outline

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- 3 Simpson's $1/3$ Rule
- 4 Composite Rule

Numerical Integration Methods

- Definite Integration problem: Find the value of $\int_a^b f(x) dx$
- Strip method
 - Divide the interval $[a, b]$ into n strips. Each strip has length $\frac{b-a}{n}$
 - Some rules in Strip Method: Rectangle Rule, Trapezoidal Rule, Midpoint Rule.
- Newton-Cotes method
 - We approximate the integration of any $f(x)$ by integration of a polynomial function $p_n(x)$:

$$\int_a^b f(x) dx \approx \int_a^b p_n(x) dx$$

- Some rules in Newton-Cotes Method: Trapezoidal Rule, Simpson's 1/3 Rule, Simpson's 3/8 Rule.

Trapezoidal rule

The trapezoidal rule reads

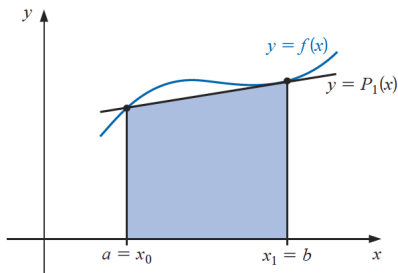
$$\int_a^b f(x) \, dx = \frac{h}{2}[f(x_0) + f(x_1)] - \underbrace{\frac{h^3}{12}f''(\xi)}_{\text{error term}} \quad (1)$$

where $h = b - a$ and $\xi \in (a, b)$. The right hand side of the formula above represents the approximation **error**. Thus we can also write

$$\int_a^b f(x) \, dx \approx \frac{h}{2}[f(x_0) + f(x_1)]. \quad (2)$$

Trapezoidal rule

This is called the Trapezoidal rule because when f is a function with positive values, $\int_a^b f(x) dx$ is approximated by the area in a trapezoid, as shown in the figure below.



Trapezoidal rule

The error term for the Trapezoidal rule involves f'' . This means that the Formula (2) gives the exact result when applied to any function whose second derivative is identically zero (i.e., any polynomial of degree one or less).

The Simpson's (1/3) rule

The Simpson's rule is defined as follows.

$$\int_a^b f(x) \, dx = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)] - \underbrace{\frac{h^5}{90}f^{(4)}(\xi)}_{\text{error term}}. \quad (3)$$

Ignoring the error term, we write

$$\int_a^b f(x) \, dx \approx \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)], \quad (4)$$

with $h = \frac{b-a}{2}$.

The Simpson's (1/3) rule

The error term in Simpson's rule involves the fourth derivative of f (i.e., $f^{(4)}$), so it gives exact results when applied to any polynomial of degree three or less.

Exercise

Compare the Trapezoidal rule (2) and Simpson's rule (4) approximations to $\int_0^2 f(x)dx$ when $f(x)$ is

① x^2

② x^4

③ $(x+1)^{-1}$

④ $\sqrt{1+x^2}$

⑤ $\sin x$

⑥ e^x

- 1 • Trapezoid:

$$\int_a^b f(x) \, dx = \frac{h}{2}(f(x_0) + f(x_1)) - \underbrace{\frac{h^3}{12}f''(\xi)}_{\text{error term}}$$

$$\begin{aligned}\int_a^b f(x) \, dx &\approx \frac{h}{2}(f(x_0) + f(x_1)) \\ &\approx \frac{2}{2}[f(0) + f(2)] \\ &\approx 4.000\end{aligned}$$

- ① • Simpson's

$$\int_a^b f(x) dx = \frac{h}{3}(f(x_0) + 4f(x_1) + f(x_2)) - \underbrace{\frac{h^5}{90}f^{(4)}(\xi)}_{\text{error term}}$$

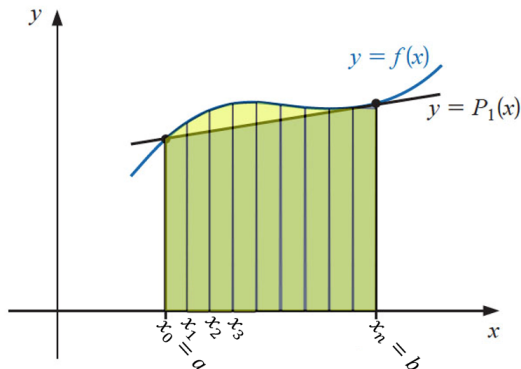
$$\begin{aligned}\int_a^b f(x) dx &\approx \frac{h}{3}(f(x_0) + 4f(x_1) + f(x_2)) \\ &\approx \frac{1}{3}[f(0) + 4f(1) + f(2)] \\ &\approx 2.667\end{aligned}$$

- ② ... continue yourself

Solution

$f(x)$	(a) x^2	(b) x^4	(c) $(x+1)^{-1}$	(d) $\sqrt{1+x^2}$	(e) $\sin x$	(f) e^x
Exact value	2.667	6.400	1.099	2.958	1.416	6.389
Trapezoidal	4.000	16.000	1.333	3.326	0.909	8.389
Simpson's	2.667	6.667	1.111	2.964	1.425	6.421

Composite Trapezoidal Rule



- Divide the interval $[a, b]$ by n :
 - $h = \frac{b-a}{n}$
 - $x_i = x_0 + hi = a + hi$

Composite Trapezoidal Rule

- Composite Trapezoidal Rule:

$$\begin{aligned}\int_a^b f(x) dx &= \int_{x_0}^{x_1} f(x) dx + \int_{x_1}^{x_2} f(x) dx + \dots + \int_{x_{n-1}}^{x_n} f(x) dx \\ &\approx \frac{h}{2}[f(x_0) + f(x_1)] + \frac{h}{2}[f(x_1) + f(x_2)] + \dots \\ &\quad + \frac{h}{2}[f(x_{n-1}) + f(x_n)] \\ &\approx \frac{h}{2} \left(f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right)\end{aligned}$$

Composite Simpson's 1/3 Rule

- Composite Simpson's 1/3 Rule:

$$\begin{aligned}\int_a^b f(x) dx &= \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots + \int_{x_{n-2}}^{x_n} f(x) dx \\ &\approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \\ &\quad + \frac{h}{3} [f(x_2) + 4f(x_3) + f(x_4)] + \dots \\ &\quad + \frac{h}{3} [f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)] \\ &\approx \frac{h}{3} \left(f(x_0) + 4 \sum_{i=1,3,5}^{n-1} f(x_i) + 2 \sum_{i=2,4,6}^{n-2} f(x_i) + f(x_n) \right)\end{aligned}$$

- Note: n must be even

Exercise

Use the Composite Trapezoidal rule and the Composite Simpson's 1/3 rule approximations to $\int_0^2 f(x)dx$ when $f(x)$ is (Note: use $n = 4$)

- 1 x^2
- 2 x^4
- 3 $(x + 1)^{-1}$
- 4 $\sqrt{1 + x^2}$
- 5 $\sin x$
- 6 e^x

"Numerical analysis" by Richard L. Burden and J. Douglas Faires

"Numerical analysis" by L. Ridgway Scott